DNS Operations update for operators who don't really want to do DNS

You know you like being called an Operator.

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DNS Operations update for operators who don't really want to do DNS

But you know, it’s one of those services that you need if you want to go on the internet. So you need a resolving name server and perhaps if you have your own domain then your going to want to get an authoritative name server as well. Am I rambling now? Has anyone read to the end of this? If you have, I appreciate it. Some terms and conditions apply.

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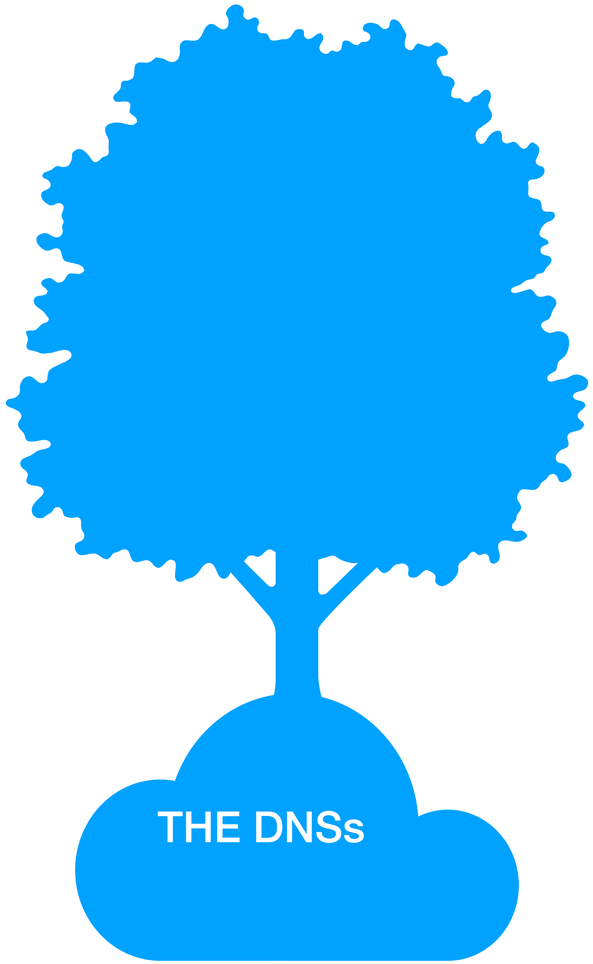

A Quick~ish refresher

Just so we are using the same terminology.

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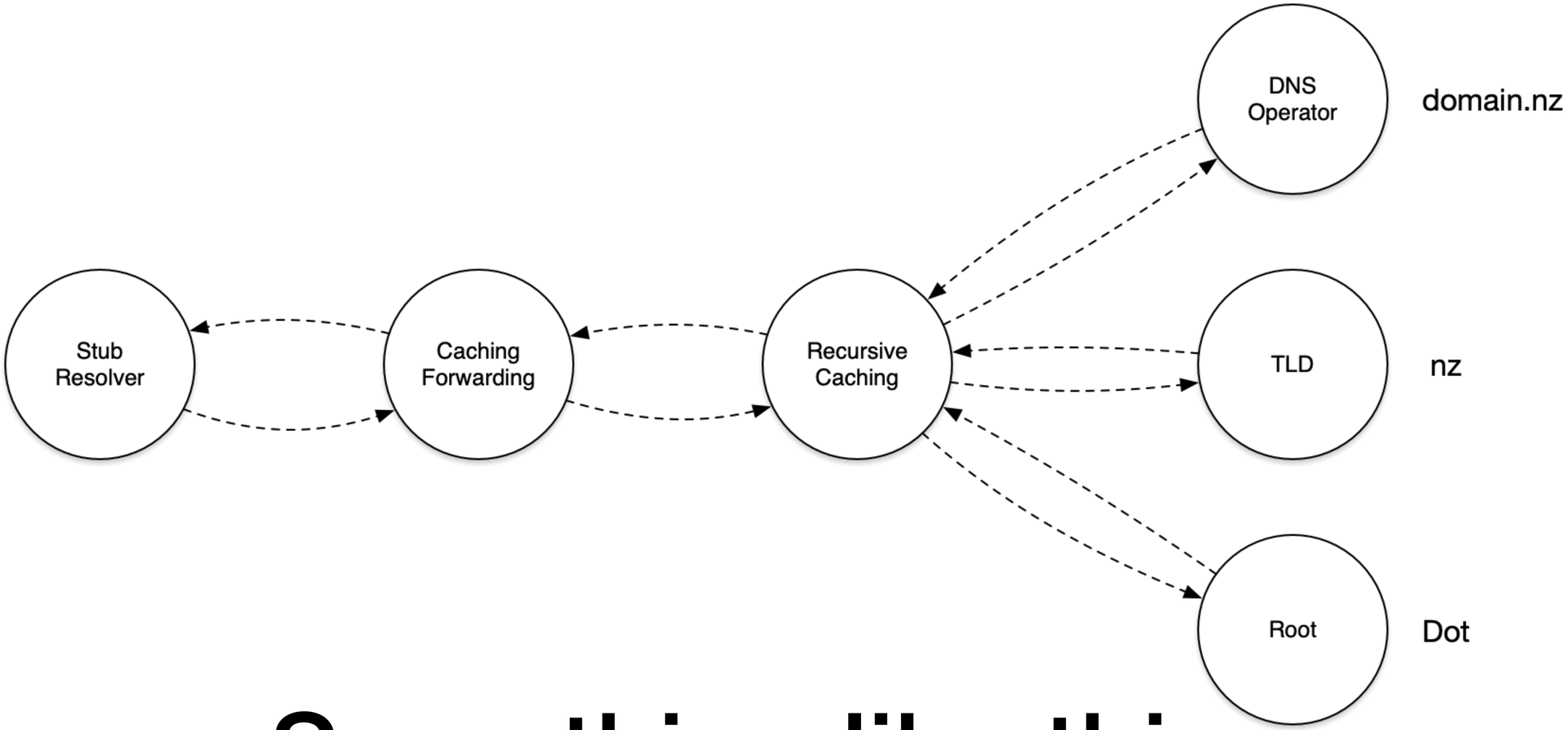

Something like this.

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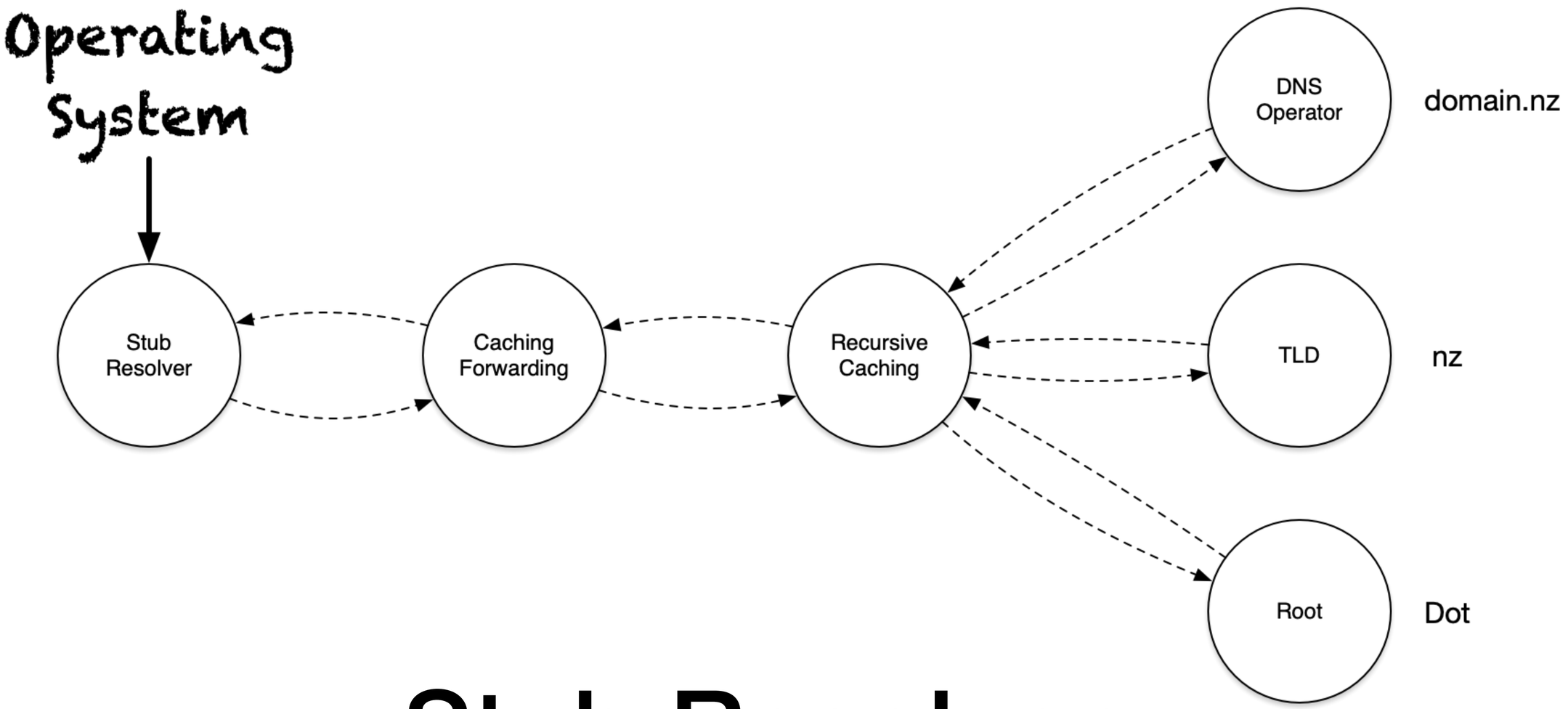

Stub Resolver

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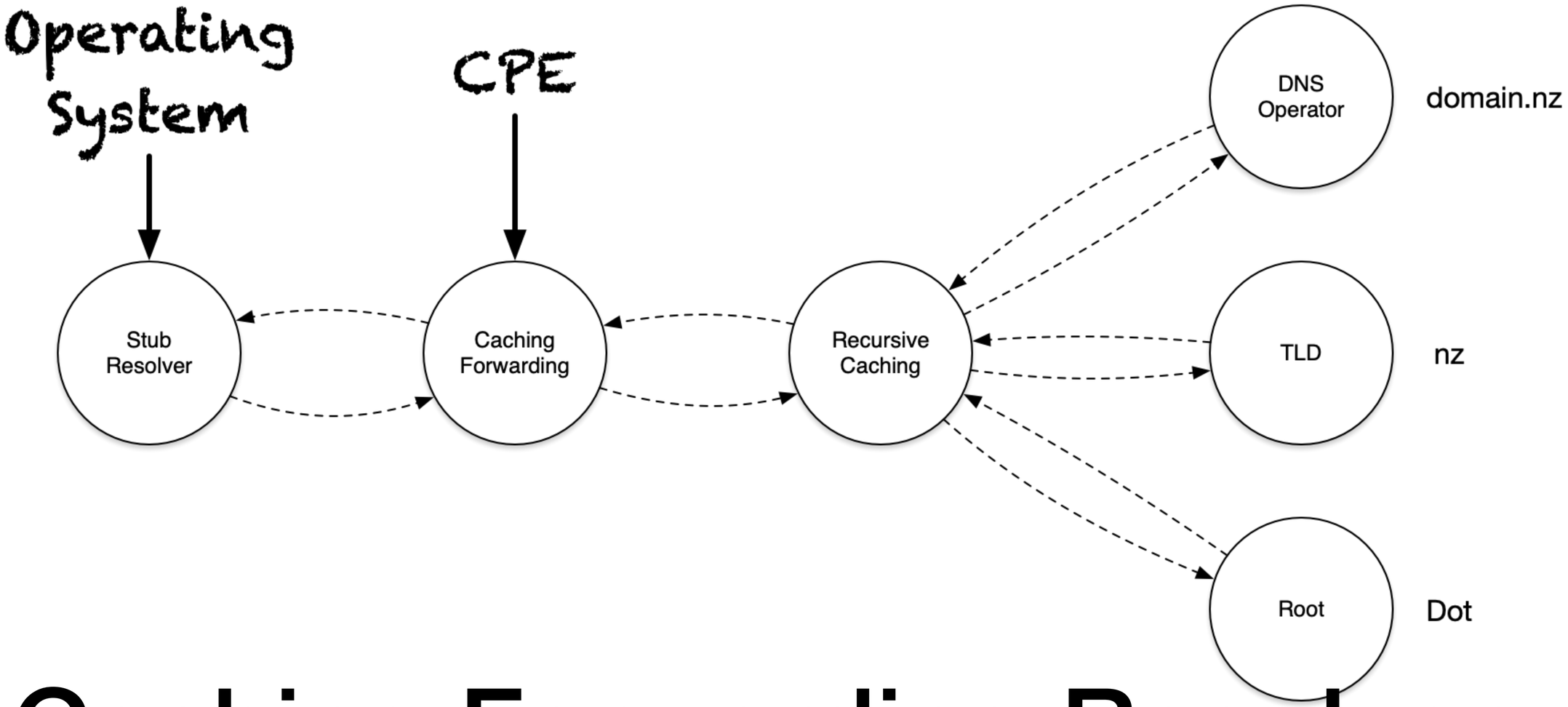
Stub Resolver

- Ask a question get an answer.
- Examples:
  - glibc (provided by OS, used by applications)
  - Whatever windows uses.
  - dig (excluding trace mode), host, etc…
- Can be security aware.



Caching Forwarding Resolver

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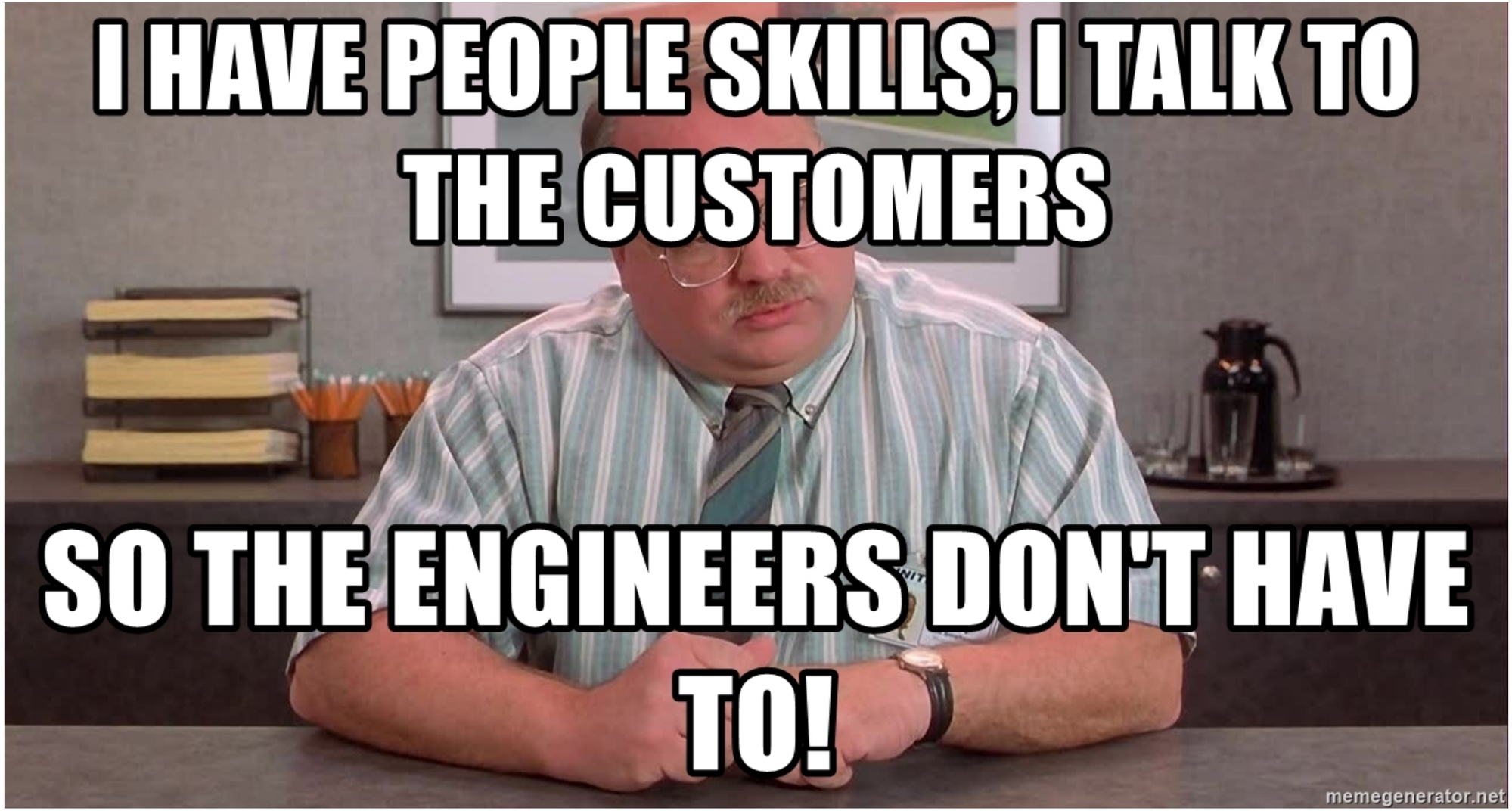

Caching Forwarder

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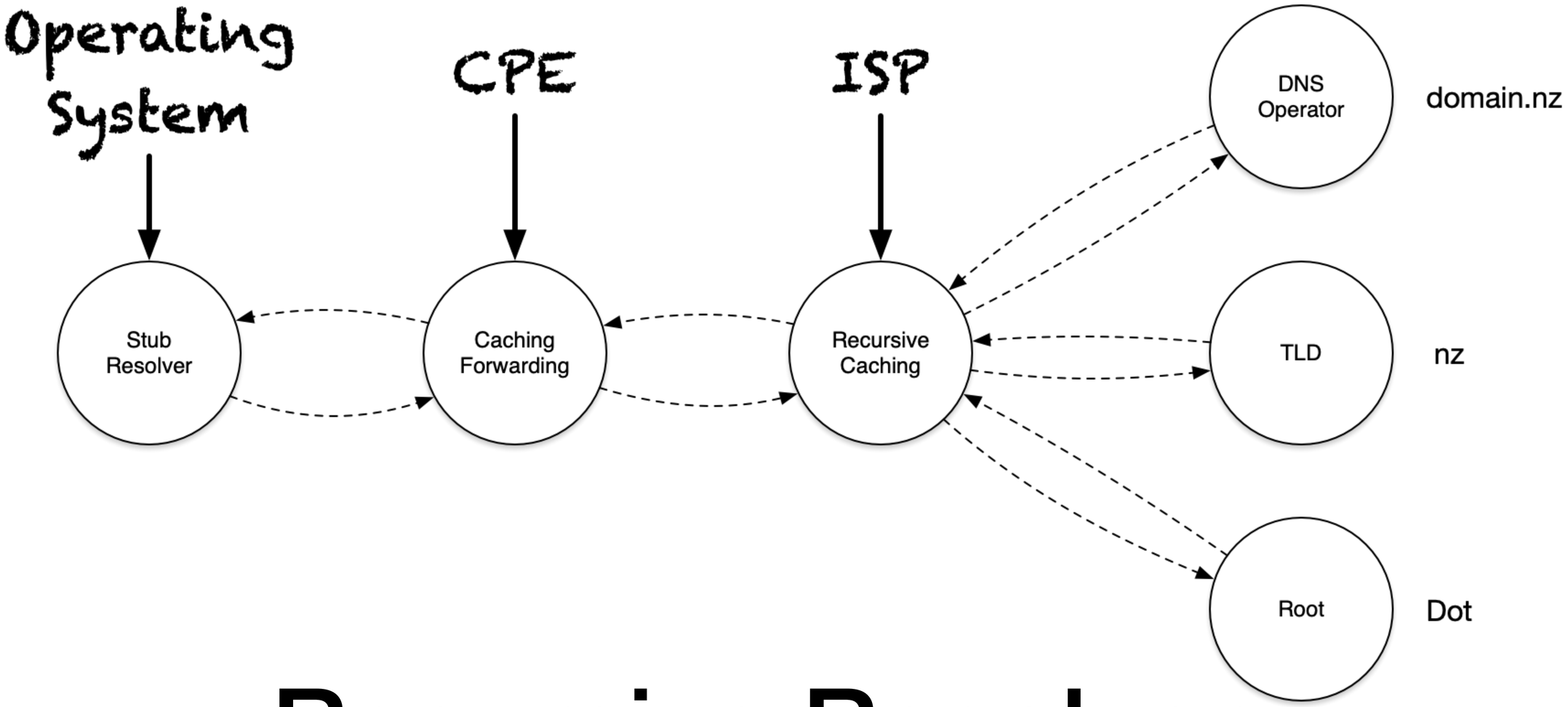
Caching Forwarder

- Takes a question and forwards it to someone else.
- Examples;
  - dnsmasq
  - Whatever garbage is in home routers.
  - systemd-resolved (sorta)
- Can be security aware.



Recursive Resolver

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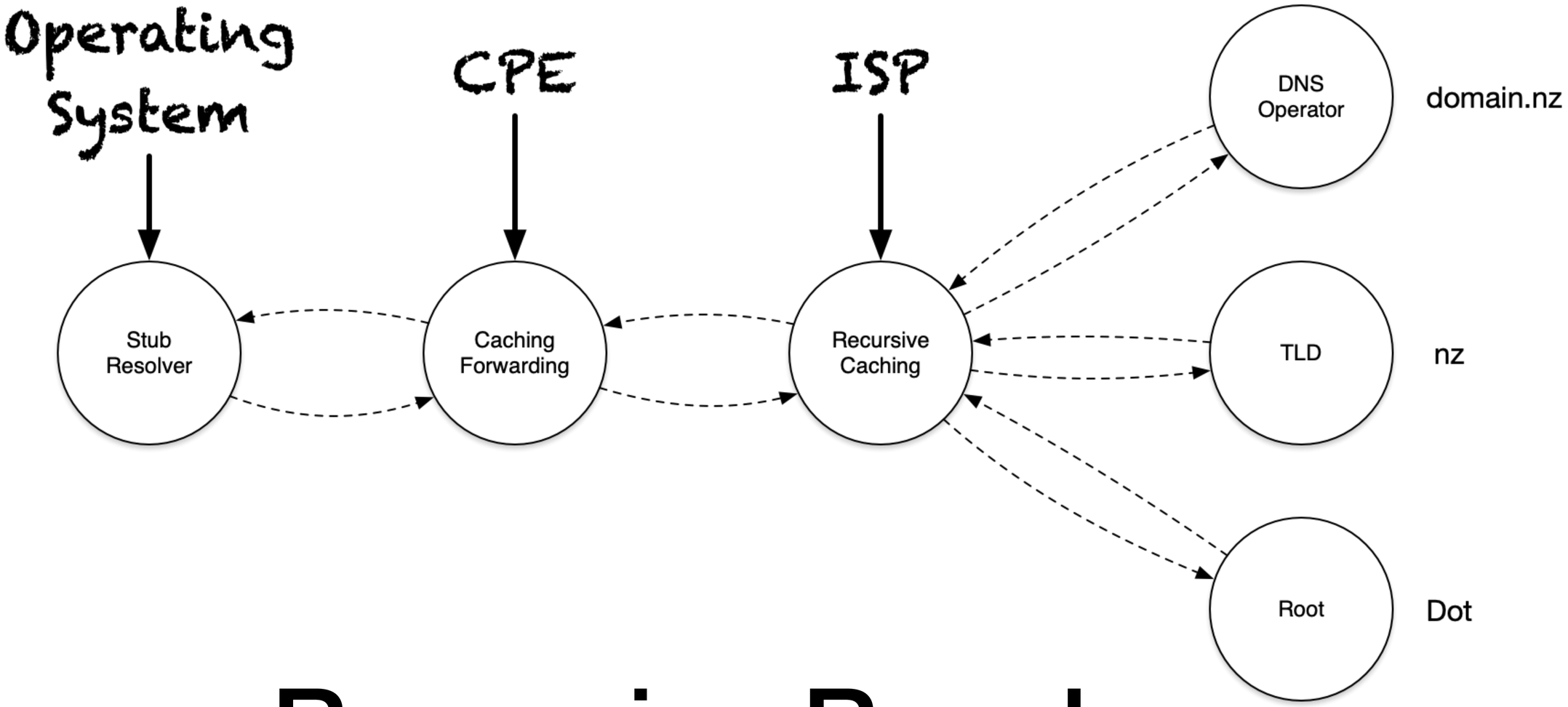
Recursive Resolver

- Accept questions and searches out the results.
- Examples;
  - Bind
  - Unbound
  - PowerDNS Recursor
  - Knot Resolver
  - Microsoft DNS
- Should be security validating.



Recursive Resolver

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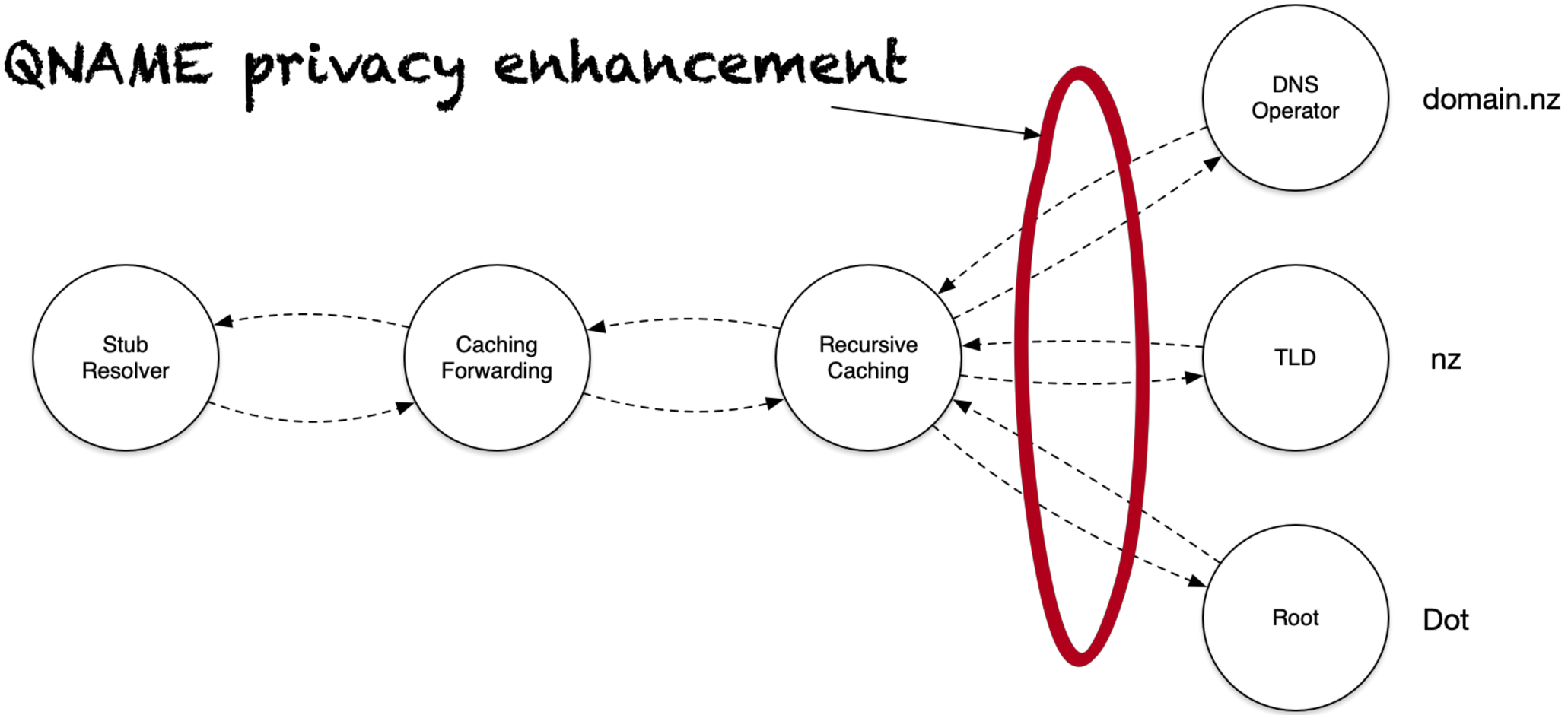
QNAME minimisation



QNAME

RFC 7816

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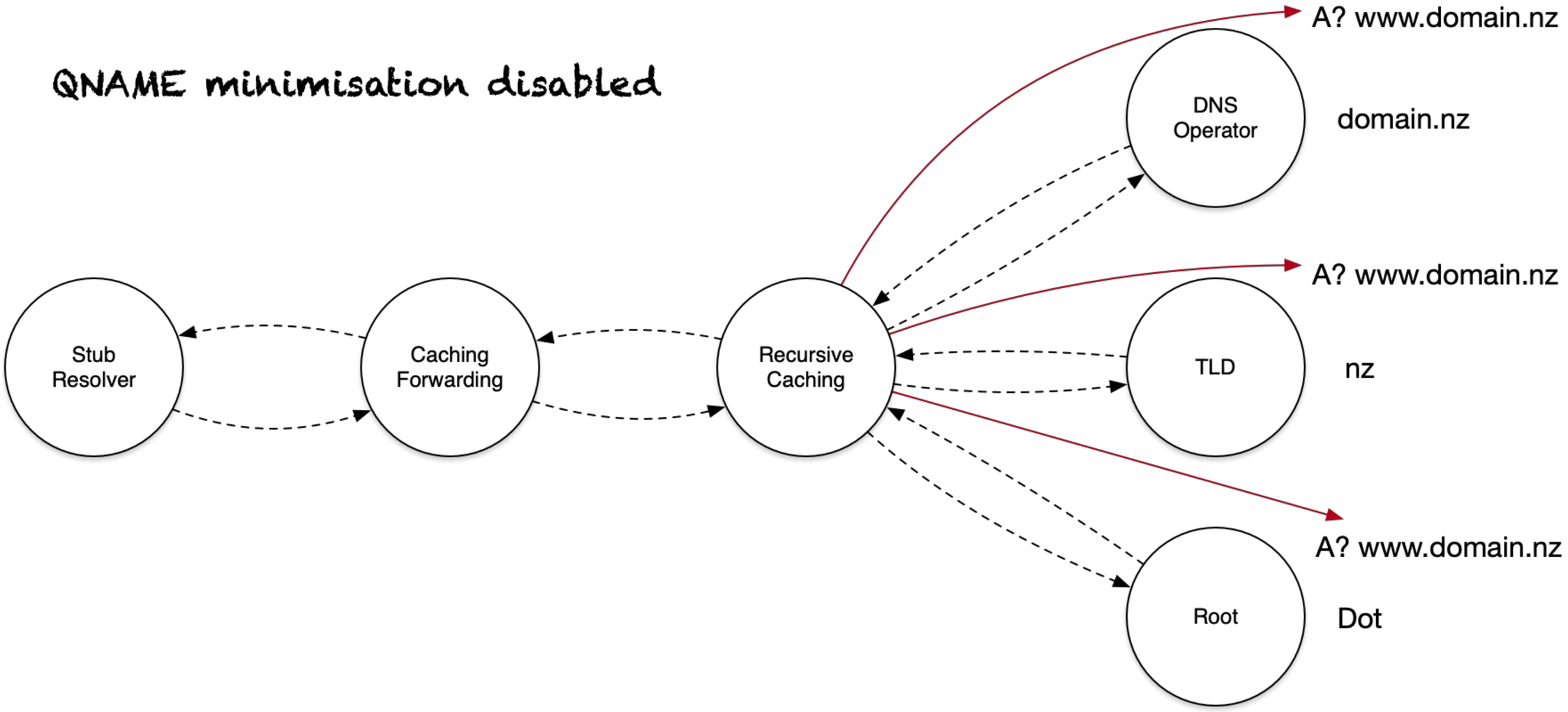
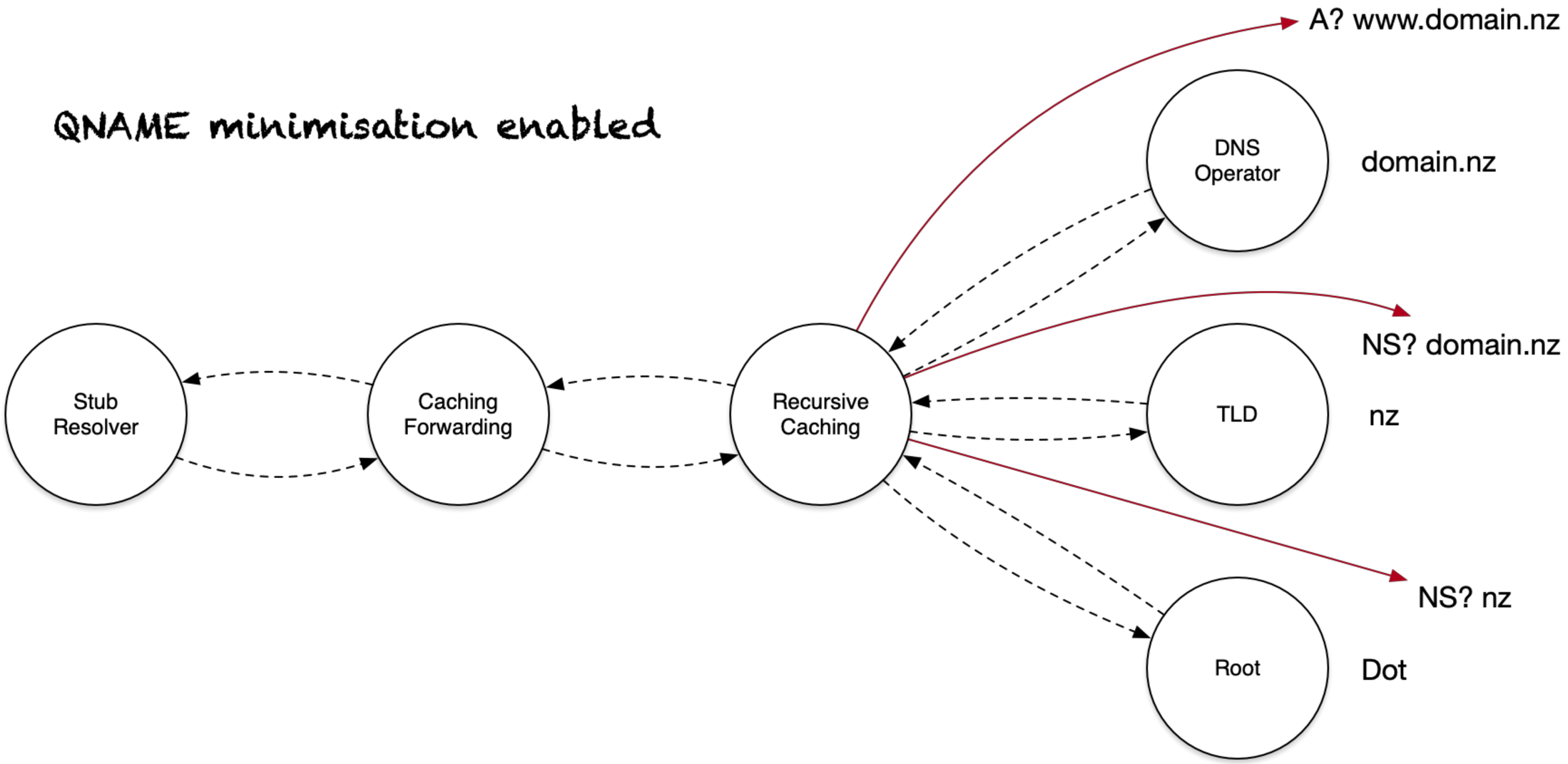
QNAME minimisation disabled



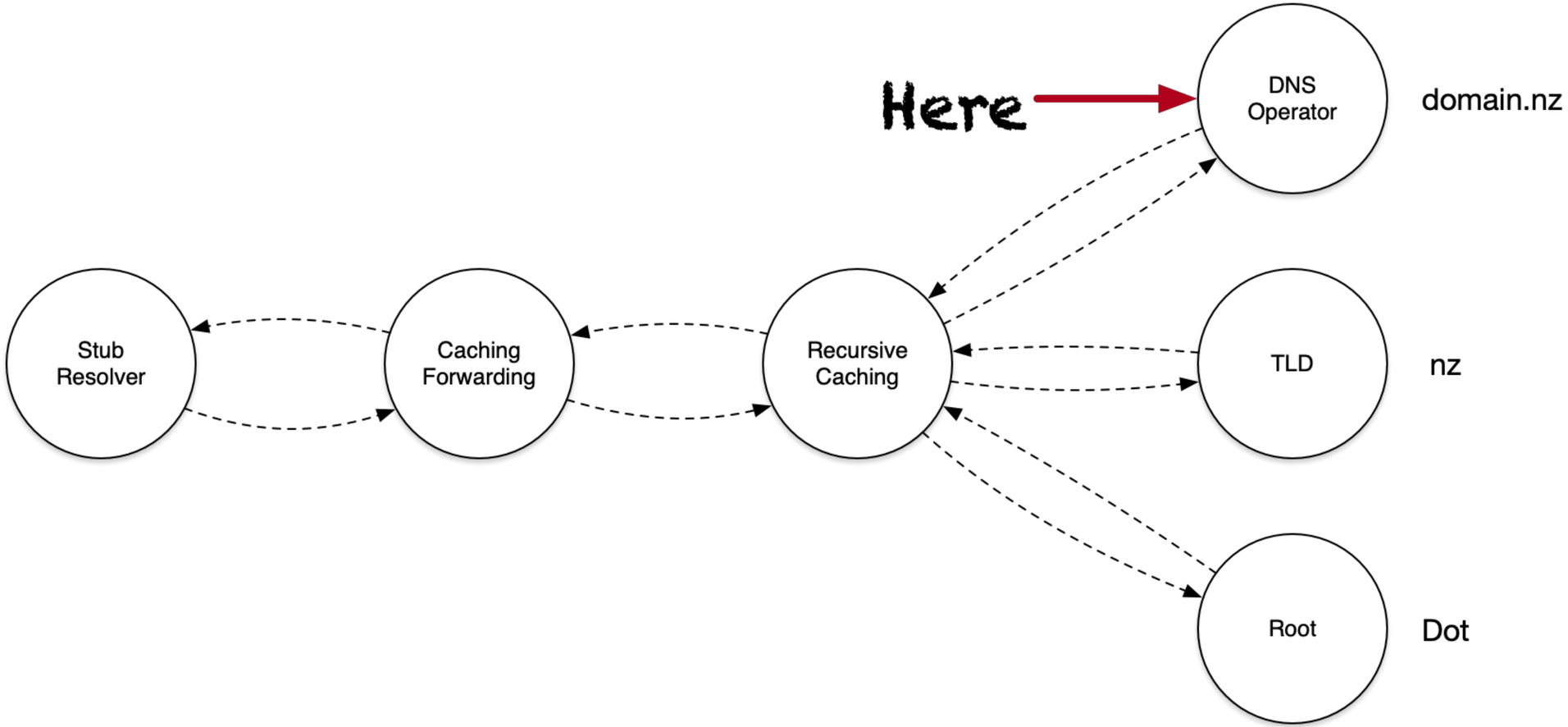
QNAME minimisation enabled



QNAME Minimisation Software Support

- Unbound (1.5.7)
- Knot Resolver (1.0)
- Bind (coming soon)

Certificate Authority Authorisation



CAA area of influence

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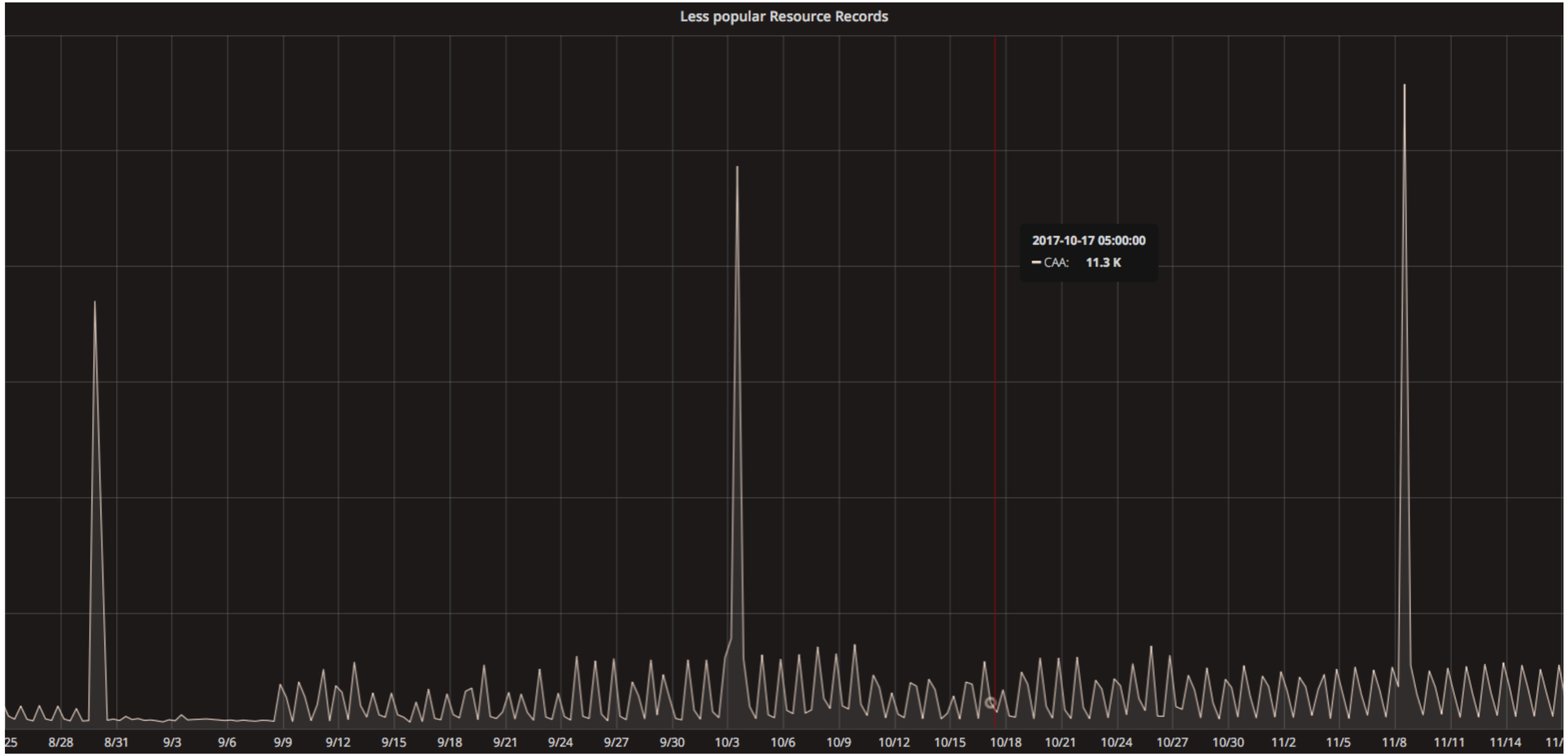
Certificate Authority Authorisation

- This allows you to publish a list of CA’s that are authorised to mint certificates for that domain.
- Mandatory



Certificate Authority Authorisation

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CAA Records Example

```
domain.nz.  IN  CAA 0 issue “letsencrypt.org”
# Only that CA may issue.

domain.nz.  IN  CAA 0 issuewild ";"
# Deny all wildcard requests.

domain.nz.  IN  CAA 0 iodef “mailto:security@domain.nz"
domain.nz.  IN  CAA 0 iodef “http://iodef.domain.nz/“
# Provide POC for security incidence.
```

https://sslmate.com/caa/

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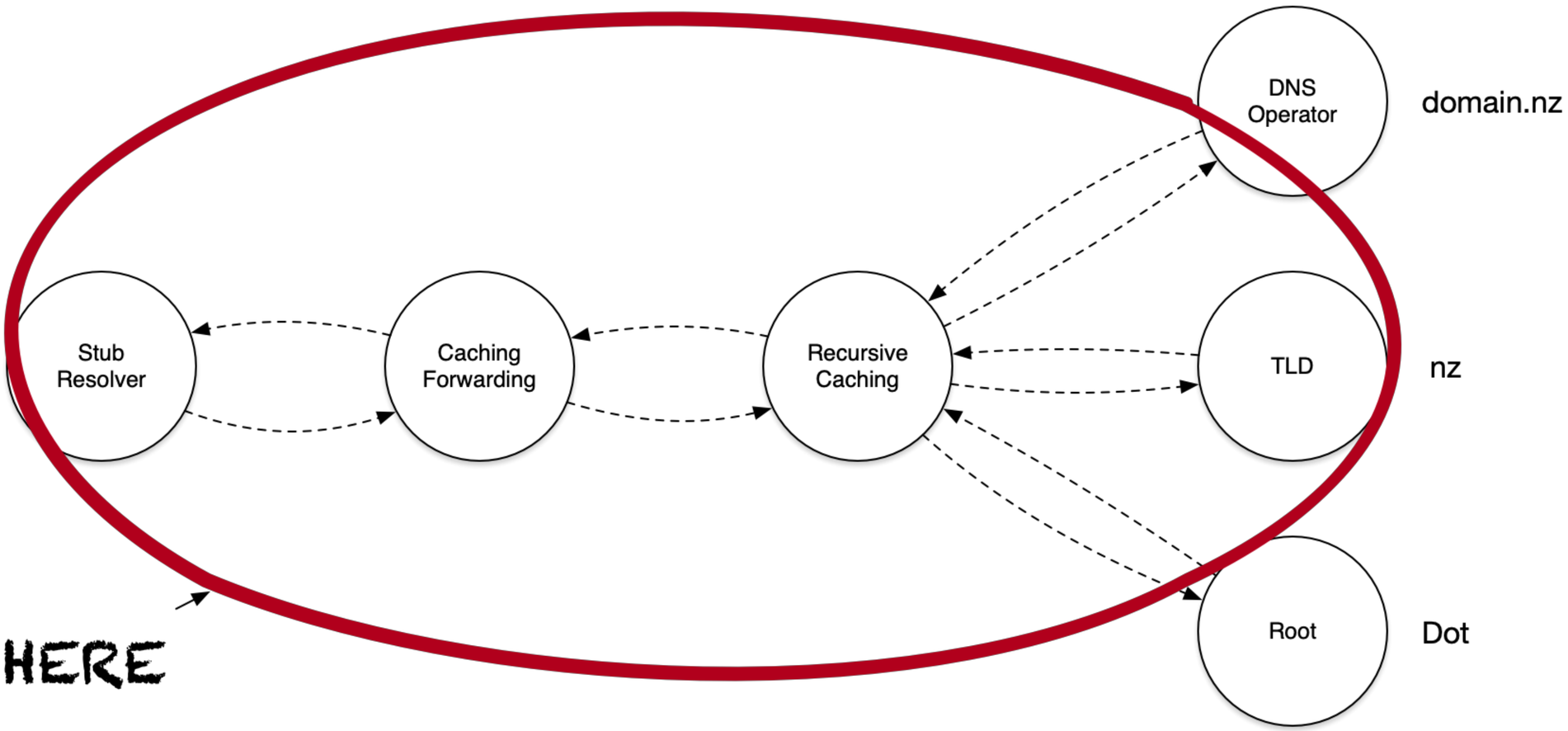
DNS Security



DNS Security

Verifiability of results.

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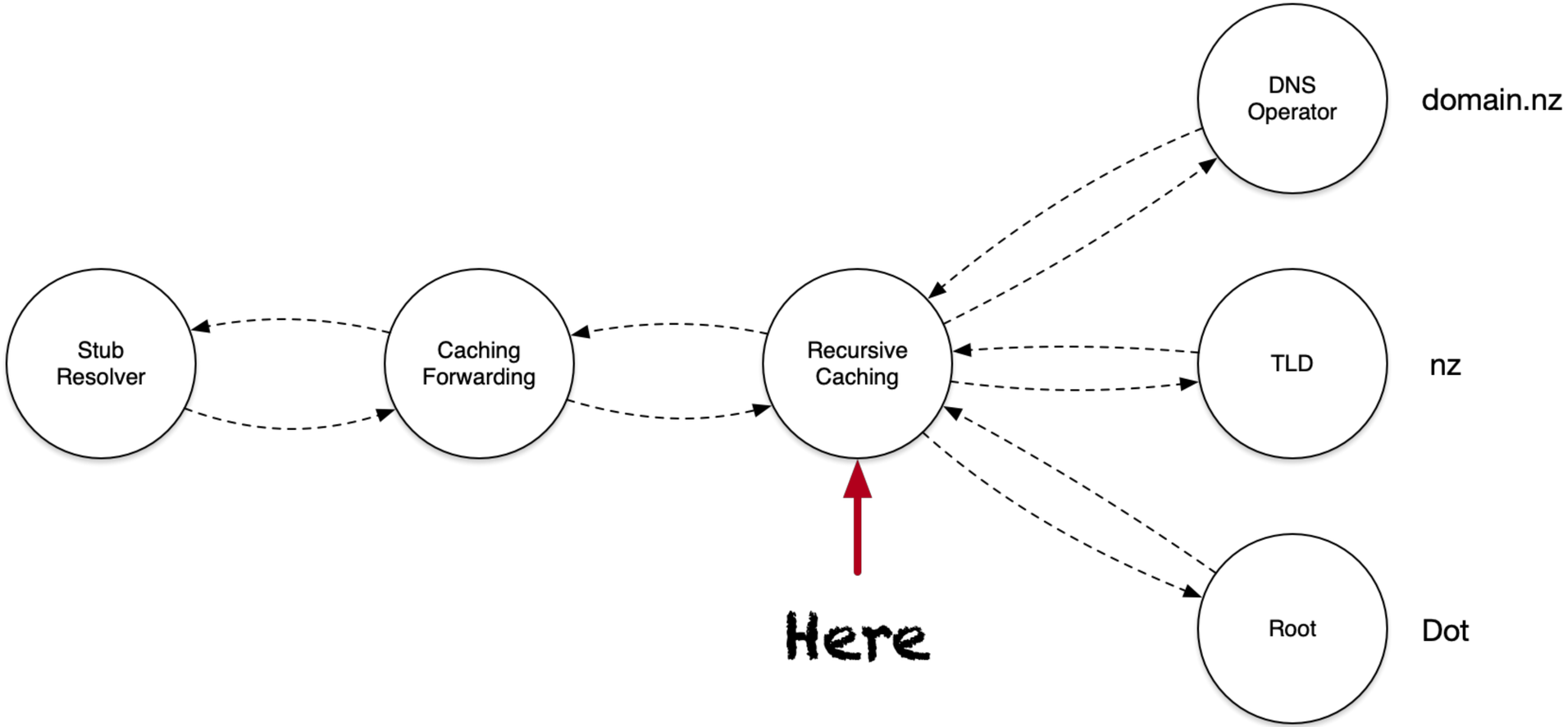
DNS Security

- DNSSEC is;
  - A system for cryptographically verifying DNS data.
  - Useful.
- DNSSEC is not;
  - Anything to do with privacy.
  - Panacea.



Recursive Deployment.

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Validating ISPs in New Zealand

- Megatel
- NowNew
- Trustpower
- Spark
- Devoli
- HDNet
- Callplus
- Compass
- Inspire
- Snap
- Flip
- Primo Wireless
- Full Flavour Media

*** Over 90% validating as per APNIC labs research.**

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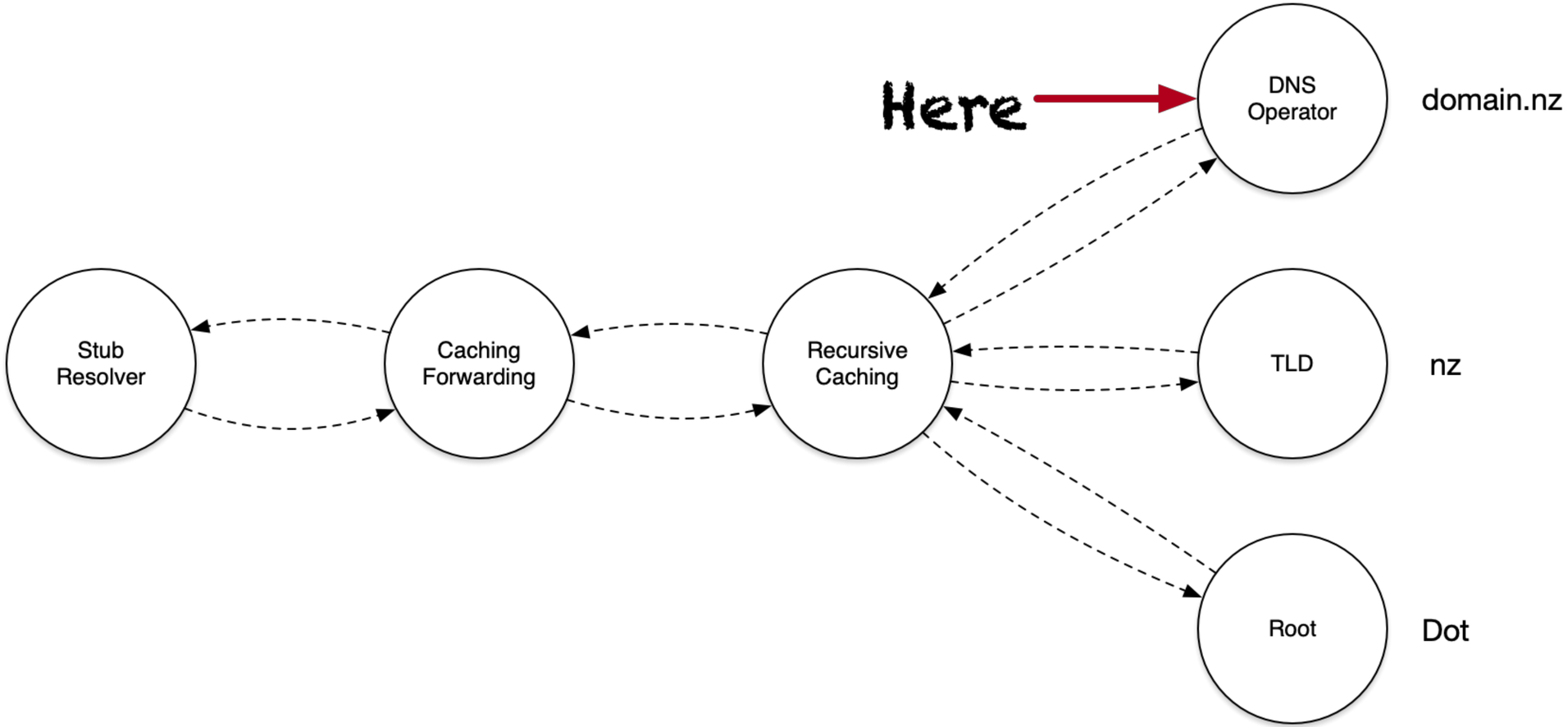
Validating Recursive Software.

- Microsoft DNS - Windows 2012.
- Bind 9.7 (2012)
- DNSmasq (2014, 2018 works)
- Unbound 0.5 (2007)
- PowerDNS Recursor 4.0 (2016)
- Knot Resolver 1.0 (2016)



Authoritative Space.

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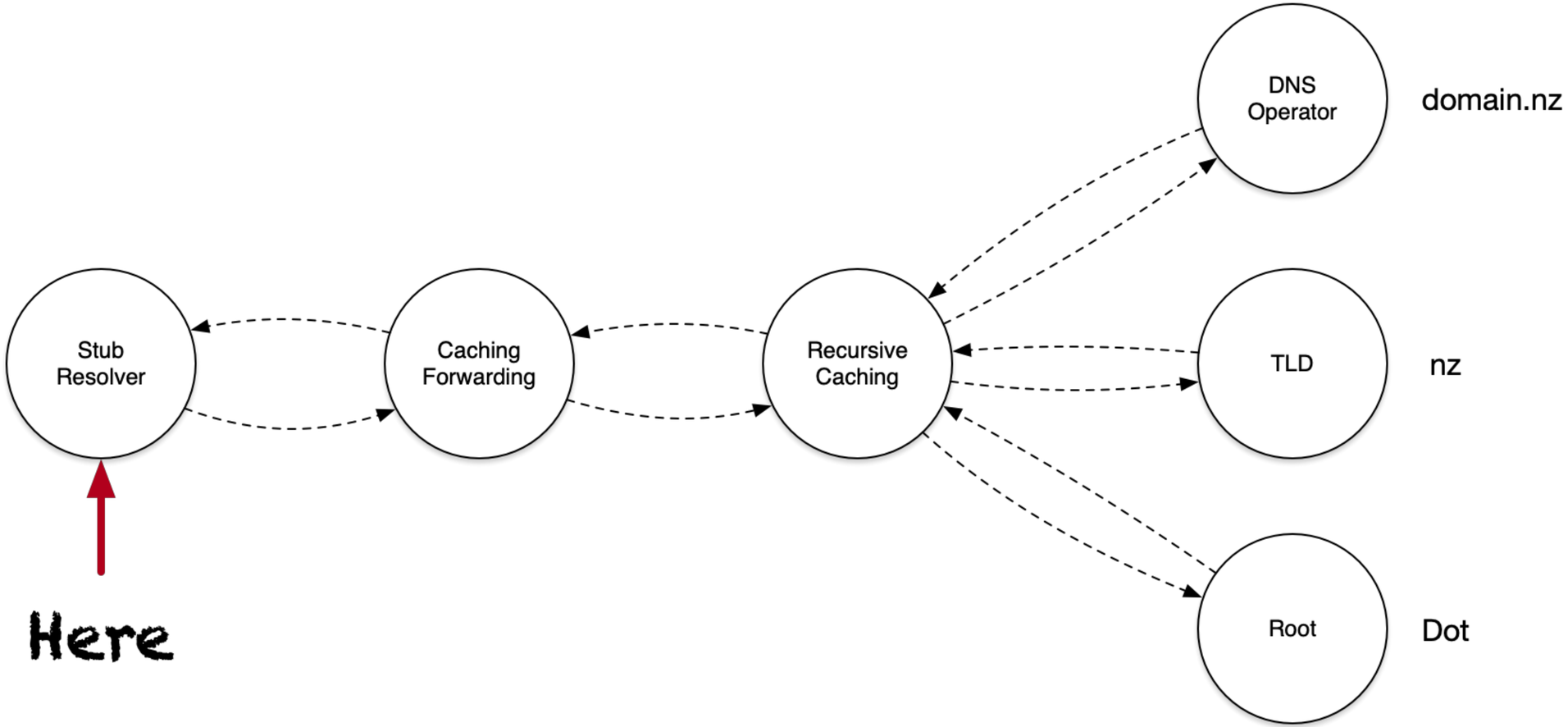
Signing Software

- Bind.
- Knot Authoritative.
- OpenDNSSec.
- DNSSEC Tools.
- Infoblox.
- Secure64.
- Microsoft DNS



Stub Resolver Space.

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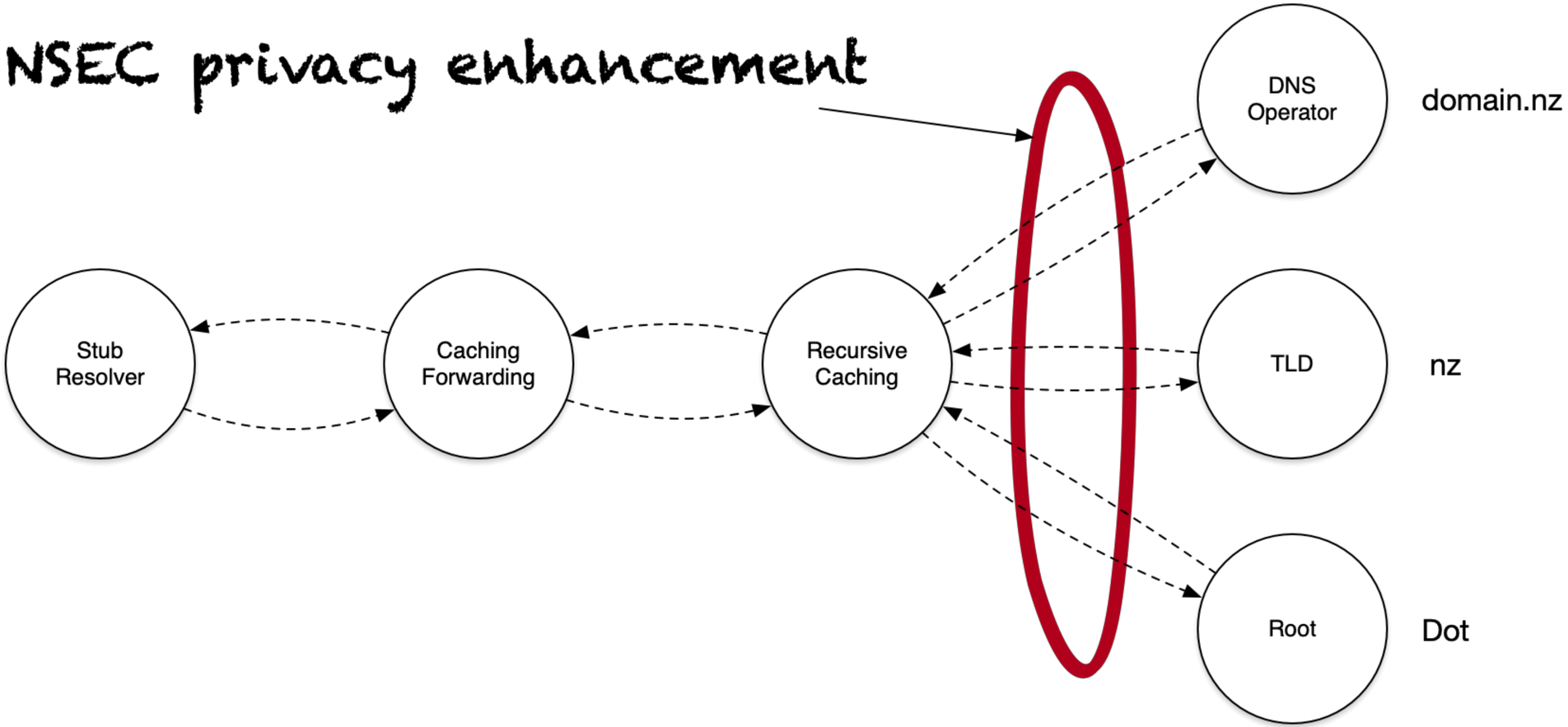
Supporting Clients

- Windows 7+
- systemd-resolverd (validating)
- glibc (aware)
- Stubby (dnsprivacy.org)

Aggressive NSEC Cache



NSEC Aggressive Caching

Privacy improvements, scaling assistance.

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example.nz
example.nz -> smtp.example.nz NSEC
smtp.example.nz
smtp.example.nz -> www.example.nz NSEC
www.example.nz
www.example.nz -> example.nz NSEC

NSEC and NSEC3

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RFC 8198

Aggressive Use of DNSSEC-Validated Cache

- Unbound
- Bind
- Knot Resolver

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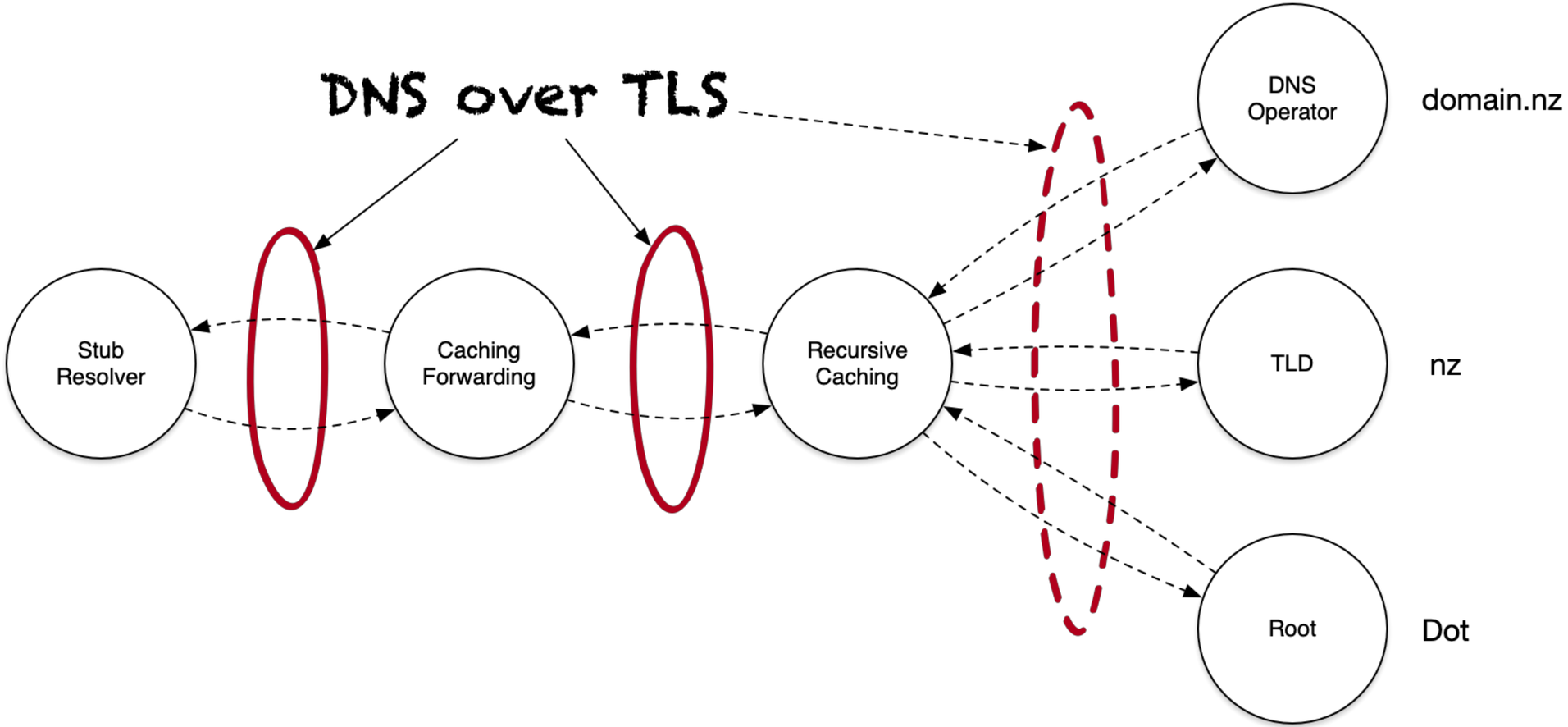
DNS over TLS



DNS over TLS

Privacy improvements.

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The Bullet Points

- Overview
  - Port 853/TCP.
  - Configuration takes an IP.
  - Can be opportunistic or strict.
- The easy stuff
  - DNS over TCP using TLS.
- The hard stuff
  - TLS credentials still a challenge.

Supporting Clients

- Android 9
  - On by default, opportunistic.
- Stubby (https://dnsprivacy.org/)
  - Third party local forwarder.
  - Support Linux, Windows and macOS.
- systemd-resolved
  - Since version 239 (pretty recent)

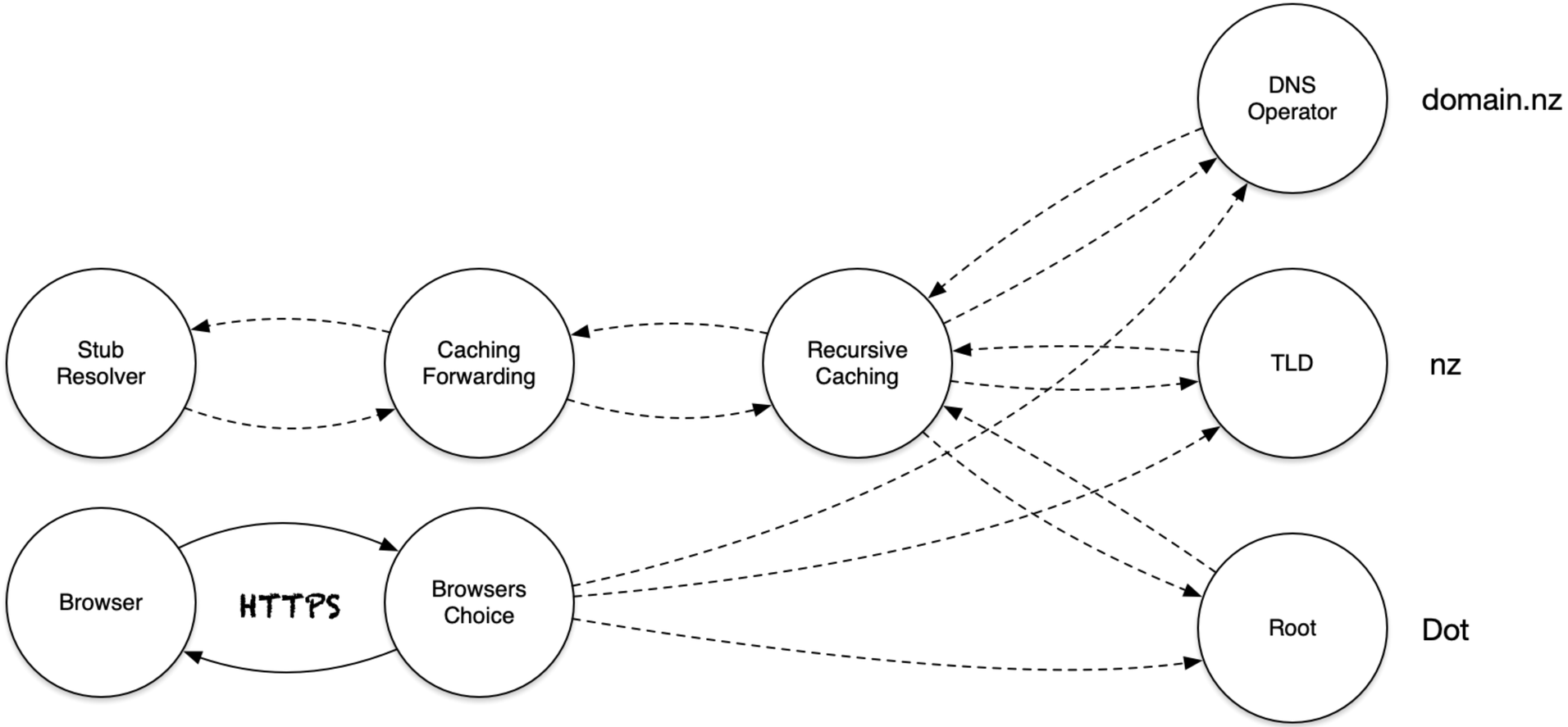
Supporting Resolvers

- Unbound.
- Knot Resolver.
- DNSdist.
- Run a TLS proxy (NGINX, haproxy).

Supporting Cloud Services

- 1.1.1.1 (Cloudflare)
- 9.9.9.9 (IBM, PCH, GCA)

DNS over HTTPS



DNS over HTTPS

Privacy improvements?

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The Bullet Points

- Overview
  - Turns DNS into a REST API.
  - Requires HTTP/2
  - Encodes DNS packets into base64.
- The easy stuff
  - It’s HTTPS, you can use the CA system.
- The hard stuff
  - It uses the CA system or certificate pinning.
  - Discovery, uses URI for configuration.

Supporting Clients

- Firefox
- Chrome
- Bromite
- Curl <- yes really!
- Proxies (Cloudflare, dnscrypt-proxy, Intra)

Supporting Resolvers

- DNScrypt
- DNSdist.
- A Lua script running in NGINX.
- Knot (soon, but Knot yet)

Supporting Cloud Services

- 1.1.1.1 (Cloudflare)
- 8.8.8.8 (Google)

DNS over JSON over HTTPS protocol.

Google implemented their own version.

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What keeps me up at night.

- Browser vendors are implementing DoH because,
  - OS hasn’t implemented it.
  - They care about user privacy.
  - Performance (no looping out via the OS).
- Why DoH?
  - HTTP Push
  - Just works if your world is browsers.
- Operators losing control of their networks (?)
  - EU vs USA, legality of consent and privacy.

Conclusions

What should I be doing?

Maintain you network services.

Use modern DNS software.

Turn on DNSSEC validation.

“Just do it.”

- Shia LaBeouf, Actual Cannibal

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Start thinking about user privacy.

Questions?